

Software Engineering

Synopsis/Project Proposal

Project Title: Blood Bank Management System | Group Number: 9

Date: 1st september 2026

Team members

- Shreelakshmi G Bhat – SRN: PES2UG24CS478
- Sangadala Yeasaswani – SRN: PES2UG24CS441
- Shreyash M Yadwad – SRN: PES2UG24CS493
- Shreenandan Jamadar – SRN: PES2UG24CS479

Proposed project description

We propose to build the Blood Bank Management System (BBMS), a new, self-contained web application rather than an extension of an existing sample project. BBMS digitizes the day-to-day operations of a blood bank - donor registration and eligibility tracking, live blood inventory management, and blood request handling between hospitals and blood banks - replacing manual, register-based record keeping. We do not yet have a committed external customer; our target users are donors (the general public), blood bank staff, hospital staff raising blood requests, and administrators who oversee accounts and reports. At the end of the project, users will be able to:

- Register as a donor and have the system check age/weight eligibility and the 90-day gap since the last donation
- View real-time blood stock levels by blood group and location, with automatic expiry flagging and low-stock alerts
- Search for available blood and submit emergency or scheduled requests, which staff can approve, reject, or fulfil
- Schedule and register for donation camps, with inventory updated automatically after each camp
- Receive SMS/email notifications for eligibility, request status, and emergency blood requirements
- View reports and an analytics dashboard summarising donations, stock, and request history

Plan of work and product ownership:

Short-term plan of work (next few weeks):

- Week 1: Finalize the SRS
- Weeks 2-4: set up the shared repository and CI, and design the database schema for donors, blood units, requests, and camps. Implement donor registration/authentication and the blood inventory module (add/update stock, expiry flagging).
- Weeks 5-6: Implement the blood request and fulfilment workflow, and integrate the SMS/email notification gateway.
- Week 7: Implement donation camp scheduling and the reporting dashboard, carry out integration testing and prepare the demo.

Who should do what:

- **Shreelakshmi G Bhat** – Donor registration & eligibility feature (functional): registration form, eligibility validation, and next-eligible-date tracking. Qualitative goal: OTP-based registration and eligibility check shall respond in under 2 seconds.
Reporting & analytics dashboard, donation, stock, and request reports plus the admin visual dashboard. Also responsible for audit logging of donations, requests, and inventory changes.

Qualitative goal: dashboard reports shall load within 3 seconds for the standard reporting period.

- **Sangadala Yeasaswani** – Blood inventory management feature (functional): add/update stock, expiry flagging, low-stock alerts. Qualitative goal: inventory search results shall return within 2-3 seconds.
- **Shreyash M Yadwad** – Blood request & fulfilment feature (functional): search, request, approve/reject/fulfil, cross-bank search. Qualitative goal: ensure role-based, secure access to approval actions.
- **Shreenandan Jamadar** – Donation camp scheduling & notifications feature (functional): camp creation/sign-up and SMS/email alerts. Qualitative goal: ensure reliable, timely delivery of eligibility and emergency notifications.